UC San Diego's Enrollment Surge and the La Jolla / University City Rental Market

Group 40

UC San Diego has expanded rapidly over the past decade and a half. Between 2010 and 2024, total enrollment increased from roughly 28,000 students to more than 45,000 students, with UCSD reporting 45,273 total students in Fall 2024 [1]. This growth can be studied using official institutional enrollment records and IPEDS fall enrollment data, which report institution-level enrollment over time [2, 3]. However, enrollment growth has not been matched by an equivalent expansion of on-campus housing capacity. As a result, many students have entered nearby rental markets, especially in University City, La Jolla, Sorrento Valley, and Pacific Beach. These neighborhoods already face strong housing demand from high-income households, biotechnology firms, coastal amenities, and limited housing supply. The central question of this project is whether UCSD enrollment growth has had a measurable effect on nearby rents, and if so, how large that effect is relative to broader San Diego County rental-market trends.

This question is important for both local and general policy reasons. Locally, rising rents near UCSD affect students, university staff, long-term residents, and lower-income renters who compete in the same housing market. More broadly, UC campuses have expanded enrollment under state pressure to increase access to higher education, while California continues to face a serious housing shortage and high housing costs, especially in coastal communities [4, 5]. If enrollment growth produces spillover effects on surrounding rental markets, then campus expansion policies should be evaluated together with housing policy. A careful empirical estimate can help distinguish the UCSD-specific effect from other forces, such as remote-work migration, low interest rates during much of the 2010s, and California's broader housing supply constraints.

The project will use three publicly available data sources. UCSD enrollment data will be obtained from the Integrated Postsecondary Education Data System maintained by the National Center for Education Statistics. IPEDS provides institution-level fall enrollment counts by year, including undergraduate and graduate enrollment [3]. ZIP code rent data will come primarily from the Zillow Observed Rent Index, which measures changes in asking rents while controlling for changes in the quality of available rental stock [6]. Apartment List Rent Estimates will be used as a secondary source to check whether the main conclusions are sensitive to the rent series used; Apartment List describes its rent estimates as combining Census rental data with real-time marketplace data [7]. ZIP-level demographic and housing controls will be drawn from the American Community Survey 5-Year Estimates through the Census data portal. These controls will include median household income, renter-occupied housing share, population density, and housing-unit characteristics [8].

The unit of analysis will be the ZIP-month. The main treated units will be ZIP codes that plausibly serve UCSD students, including 92122, 92037, 92121, and 92109. The comparison group will consist of San Diego County ZIP codes located farther from campus and matched on pre-period income levels, renter share, and housing density. The main empirical design will be a panel difference-in-differences model with ZIP fixed effects and month fixed effects. The baseline specification will be

$$\log(\text{Rent}_{zt}) = \beta\,(\text{UCSDFeeder}_z \times \text{UCSDEnrollment}_t) + \alpha_z + \gamma_t + X_{zt}\theta + \varepsilon_{zt},$$

where z indexes ZIP codes and t indexes months. The coefficient β measures whether rents in UCSD-feeder ZIP codes rise more strongly during periods of enrollment growth than rents in comparable ZIP codes farther from campus. ZIP fixed effects control for time-invariant differences across neighborhoods, while month fixed effects absorb county-wide and macroeconomic shocks. The covariate vector X_{zt} will include ACS-based demographic and housing controls, interpolated to monthly frequency where necessary.

The analysis will include several validation and robustness checks. First, I will examine whether UCSD-feeder ZIP codes and matched comparison ZIP codes show similar rent trends in the early part of the Zillow sample. Second, I will run placebo regressions using pre-2020 subperiods to test whether the estimated treatment relationship appears before the strongest recent enrollment and rent pressures. Third, I will compare results across different definitions of the treated area, including a narrower La Jolla and University City treatment group and a broader distance-based treatment group. Fourth, I will test whether smaller units, such as studios and one-bedroom apartments, respond more strongly than larger units. This subgroup analysis is important because student demand should be more directly reflected in smaller rental units. Finally, I will compare Zillow-based estimates with Apartment List estimates to assess robustness to measurement differences across rent datasets.

The expected outcome is a quantitative estimate of the relationship between UCSD enrollment growth and rent growth in nearby ZIP codes from 2014 to 2024. I expect the UCSD effect to be statistically detectable but not large enough to explain all rent appreciation in the surrounding neighborhoods. A plausible outcome is that UCSD enrollment growth explains a meaningful share of the differential rent growth in nearby student-feeder ZIP codes, while a large residual share remains attributable to region-wide housing demand, income growth, and supply constraints. The final project will report point estimates, confidence intervals, visual event-study plots, and robustness tables. It will also provide a decomposition of observed rent growth into a UCSD-associated component and a broader market component.

This project is feasible within the course timeline because the required datasets are publicly available, downloadable in CSV format, and modest in size. The empirical design is challenging enough for a small group project because it requires panel construction, treatment and control selection, fixed-effects regression, robustness checks, and interpretation of causal assumptions. At the same time, the project is realistic because it does not require restricted data, scraping, or complex machine learning infrastructure. The final analysis will directly apply statistical inference methods, including hypothesis testing, confidence intervals, regression modeling, and sensitivity analysis, to a policy-relevant local question.

Artificial intelligence tools were used only for grammar checking and language polishing. The tool helped identify wording, sentence clarity, and minor grammatical issues, but it was not used to generate the project idea, conduct the analysis, select the methods, or produce the substantive content of the proposal. All final decisions about content, structure, and interpretation were made by the authors.

References

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[2] UC San Diego Institutional Research. "Undergraduate and Graduate Enrollment Dashboards."

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[4] California Legislative Analyst's Office. "California's High Housing Costs: Causes and Consequences." 2015.

[5] University of California. "University of California Sets Enrollment Records." 2026.

[6] Zillow Research. "Methodology: Zillow Observed Rent Index."

[7] Apartment List Research. "Introducing the New Apartment List Rent Estimate Methodology."

[8] U.S. Census Bureau. "American Community Survey 5-Year Data."